Homework 1: Statistical Review, Forecast Baselines, and Agent Audit

Course: ISyE 4031
Version: Public sample
Work mode: Individual
Format: Reproducible project folder or ZIP archive

Purpose

This assignment asks you to use statistical reasoning and an AI coding agent together. The agent may propose code, explanations, and tests, but you remain responsible for the estimand, assumptions, calculations, information cutoff, and conclusions.

Your files must allow another person to reproduce every reported result without editing your code.

Required deliverables

Create a project folder containing the following files at the top level:

- `analysis.py` or a clearly organized notebook that produces every numerical result;
- `test_analysis.py` with the required automated checks;
- `report.pdf`, no more than five pages, answering all written questions;
- `AI_USAGE.md`, using the evidence requirements in Part 4;
- a short `README.md` with exact commands for reproducing the analysis and tests.

If you share the project as a ZIP archive, name it `hw1_lastname_firstname.zip`. Do not include a Git repository, virtual environment, package cache, credentials, or unrelated files.

Run the following before considering the work complete:

```
python analysis.py
pytest -q
```

Then create the optional archive. If you used a notebook, substitute its filename for `analysis.py`.

```
zip hw1_lastname_firstname.zip analysis.py test_analysis.py \
    report.pdf AI_USAGE.md README.md
```

Part 1: Descriptive summaries and uncertainty

The following course-provided practice data represent 20 consecutive daily maximum temperatures in degrees Fahrenheit. Treat the first 14 observations as the legal information set and the final 6 observations as a held-out evaluation period.

```
temps = [86, 88, 91, 90, 93, 95, 94, 92, 89, 87,
         88, 90, 92, 94, 96, 95, 93, 91, 90, 88]
train = temps[:14]
test = temps[14:]
```

Do not use the held-out values when estimating a model or interval.

1a. Plot the 20 temperatures in their observed order. Mark the cutoff after day 14, distinguish training from held-out observations, label both axes, and write a one-sentence caption explaining what information is legal when the forecasts are issued. **[4 points]**

1b. State the unit of analysis, response, information cutoff, and intended forecast target. **[4 points]**

1c. Using only `train`, calculate the mean, median, range, sample variance, sample standard deviation, and interquartile range. Show enough work to verify the denominator used for the variance. **[6 points]**

1d. Assuming the training observations are independent draws from an approximately normal population, construct a 95% t confidence interval for the population mean. State the formula and interpret the interval as a repeated-sampling procedure. **[5 points]**

1e. Under the same assumptions, construct a 95% prediction interval for one new daily maximum temperature. Explain why this interval is wider than the interval in 1d. **[6 points]**

Part 2: A transparent forecast baseline

Use the same fixed training set to create two six-step forecasts:

- **Mean baseline:** forecast every held-out day using the training mean.
- **Last-value baseline:** forecast every held-out day using the final observed training value.

2a. Report the six forecasts from each baseline and overlay both forecast paths on the ordered-data plot from Part 1. Explain why neither forecast may change after the held-out period begins. **[5 points]**

2b. Calculate mean absolute error and root mean squared error for both baselines on `test`. Identify the better baseline under each metric. **[8 points]**

2c. Explain one reason these six errors are not six independent replications of model performance. **[4 points]**

2d. Choose one baseline as the required benchmark for a future extension. Defend the choice using simplicity, performance, and reproducibility. **[5 points]**

Part 3: One-sample inference

A melting-point test of $n = 10$ binder samples produced $\bar{x} = 154.2°\text{F}$. Assume the population is normal with known standard deviation $\sigma = 1.5°\text{F}$.

3a. Test $H_0 : \mu = 155$ against $H_a : \mu \neq 155$ at $\alpha = 0.01$. Show the test statistic, rejection rule, decision, and conclusion in context. **[8 points]**

3b. Calculate and interpret the two-sided p-value. Do not describe it as the probability that H_0 is true. **[5 points]**

3c. Calculate the probability of a Type II error when the true mean is 150°F. State the acceptance region used in the calculation. **[7 points]**

3d. Verify all three results in Python and compare them with your independent calculation. **[4 points]**

Part 4: Audit an agent-generated analysis

An AI agent proposes the following code for Part 1 and Part 2:

```
import numpy as np
from scipy import stats

x = np.array(temps, dtype=float)
train = x[:14]
test = x[14:]

center = x.mean()
spread = train.std()
se = spread / len(train)
mean_interval = stats.norm.interval(.95, loc=center, scale=se)
future_interval = mean_interval
forecast = np.repeat(x[-1], len(test))
```

4a. Identify at least five distinct statistical, forecasting, or reproducibility problems. For each problem, explain the consequence rather than merely naming the faulty line. **[10 points]**

4b. Replace the code with a correct implementation. Keep the two transparent baselines separate from any later extension. **[6 points]**

4c. Add automated tests demonstrating all four properties below: **[8 points]**

- changing held-out values cannot change training summaries or forecasts;
- the sample standard deviation uses $n - 1$ in the denominator;
- the prediction interval is wider than the confidence interval for the mean;
- repeated runs produce identical results.

AI Usage Statement

Use at least one designated course agent for Part 4. Your `AI_USAGE.md` must include:

- the tool and model used;
- one representative prompt or prompt template;

- the task delegated to the agent;
- the files or functions the agent read or changed;
- the tests the agent proposed or ran;
- one suggestion you accepted, revised, or rejected, with your statistical reason;
- one unresolved limitation or risk.

Do not include private chain-of-thought or unrelated account history. A concise project-specific audit record is sufficient. **[5 points]**

Rubric summary

Component	Points
Descriptive summaries and uncertainty	25
Forecast baselines and evaluation	22
One-sample inference	24
Agent audit and automated tests	24
AI Usage Statement	5
Total	**100**

Reproducibility checks

Before considering the work complete, confirm that:

- the ZIP file opens and contains the five required files at its top level;
- all reported numbers can be regenerated from the project files;
- no held-out observation enters training, interval estimation, or baseline construction;
- every assumption and interpretation is written in your own defensible language;
- `pytest -q` passes in the course Codespace;
- no credentials, tokens, private data, or unrelated files are included.